

Problem Set 2 with Suggested Solutions

THE CHALLENGES OF GROWTH

Problem 1 Carbon Emissions and Growth. It is possible to extend the production function so that output is produced with labor input, N ; capital input, K , and carbon-intensive energy, E . If we write a production function as

$$Y = N^{1/3}K^{1/3}E^{1/3}$$

- 1) Using the input values below, does this production function exhibit constant returns to scale?

Units of N	Units of K	Units of E	Output
27	125	64	60
54	250	128	120
81	375	192	180

Answer: Note that $f(tN, tK, tE) = (tN)^{1/3}(tK)^{1/3}(tE)^{1/3} = t(N^{1/3}K^{1/3}E^{1/3}) = tf(N, K, E) = tY$ for all $t \geq 0$. Thus, the given production function exhibits constant returns to scale.

- 2) Solve for the necessary amount of energy to produce the specified level of output in each row below.

Units of N	Units of K	Units of E	Output
54	250	128	120
54	375	$\frac{2^8}{3} \approx 85.33$	120
81	250	$\frac{2^8}{3} \approx 85.33$	120

- 3) Comment on the following statement using your results from 2).

"Because of constant returns to scale, in order for output to rise, we must increase our consumption of energy."

Answer: The given statement is false. Note that the production function is increasing in each input. Therefore, output can increase if labor or/and capital increase(s) without increases in consumption of energy.

- 4) Use words to describe how output can increase without an increase in the use of energy. The text points out that there is widespread agreement among economists that a tax on carbon emissions energy is the best way to reduce carbon emissions. How does this proposition fit into the production function above?

Answer: The answer may depend on how much the carbon emission tax per unit will be levied. If the marginal cost of using emissions is less than the marginal benefit(product) of energy, an introduction of a carbon tax may not be effective. Furthermore, in the long run, technological progress through R&D and innovation seems to be a better way than a carbon tax.

FINANCIAL MARKETS AND EXPECTATIONS

Problem 1 Approximating the Price of Long-Term Bonds. *The present value of an infinite stream of dollar payments of $\$z$ (that starts next year) is $\$z/i$ when the nominal interest rate, i , is constant. This formula gives the price of a consol—a bond paying a fixed nominal payment each year, forever: It is also a good approximation for the present discounted value of a stream of constant payments over long but not infinite periods, as long as i is constant. Let's examine how close the approximation is.*

1) Suppose that $i = 10\%$. Let $\$z = 100$. What is the present value of the consol?

Answer:

$$\begin{aligned} \$PV_{consol} &= \frac{\$z}{(1+i)} + \frac{\$z}{(1+i)^2} + \cdots \\ &= \frac{\$z}{(1+i)} \frac{1}{1 - \frac{1}{(1+i)}} \\ &= \frac{\$z}{(1+i)} \frac{1}{i} \\ &= \frac{\$z}{i} = \frac{\$100}{0.1} = \$1,000. \end{aligned}$$

2) If $i = 10\%$, what is the expected present discounted value of a bond that pays $\$z$ over the next 10 years? 20 years? 30 years? 60 years? (*Hint: Use the formula from the chapter but remember to adjust for the first payment.*)

Answer: Note that for a t -year bond ,

$$\$PV_{bond} = \frac{\$z}{(1+i)} + \frac{\$z}{(1+i)^2} + \cdots + \frac{\$z}{(1+i)^t}$$

Premultiply both sides of the above equation by $\frac{1}{(1+i)}$,

$$\frac{1}{(1+i)} \$PV_{bond} = \frac{\$z}{(1+i)^2} + \frac{\$z}{(1+i)^3} + \cdots + \frac{\$z}{(1+i)^t} + \frac{\$z}{(1+i)^{t+1}}$$

Subtract above two equations side by side,

$$\begin{aligned} \left(1 - \frac{1}{(1+i)}\right) \$PV_{bond} &= \frac{\$z}{(1+i)} - \frac{\$z}{(1+i)^{t+1}} \\ \Rightarrow \frac{i}{(1+i)} \$PV_{bond} &= \frac{\$z}{(1+i)} - \frac{\$z}{(1+i)^{t+1}} \\ \Rightarrow \$PV_{bond} &= \frac{\$z}{i} - \frac{\$z}{i(1+i)^t} \end{aligned}$$

Thus, for $\$z = 100$ and $i = 0.1$,

- when $t = 10$, $\$PV_{bond} = \frac{\$100}{0.1} - \frac{\$100}{0.1(1.1)^{10}} \approx \$1,000 - \$385.54 \approx \614.46
- when $t = 20$, $\$PV_{bond} = \frac{\$100}{0.1} - \frac{\$100}{0.1(1.1)^{20}} \approx \$1,000 - \$148.64 \approx \851.36
- when $t = 30$, $\$PV_{bond} = \frac{\$100}{0.1} - \frac{\$100}{0.1(1.1)^{30}} \approx \$1,000 - \$57.31 \approx \942.69

- when $t = 60$, $PV_{bond} = \frac{\$100}{0.1} - \frac{\$100}{0.1(1.1)^{60}} \approx \$1,000 - \$3.28 \approx \996.72

Note that we can approximate the price of long-term bonds as the price of a consol when the maturity of bonds is getting longer.

- 3) Repeat the calculation in 1) and 2) for $i = 2\%$ and $i = 5\%$.

Answer:

For $i = 2\%$, $PV_{consol} = \frac{\$100}{0.02} = \$5,000$.

- when $t=10$, $PV_{bond} = \frac{\$100}{0.02} - \frac{\$100}{0.02(1.02)^{10}} \approx \$5,000 - \$4,101.74 \approx \896.26
- when $t=20$, $PV_{bond} = \frac{\$100}{0.02} - \frac{\$100}{0.02(1.02)^{20}} \approx \$5,000 - \$3,364.86 \approx \1635.14
- when $t=30$, $PV_{bond} = \frac{\$100}{0.02} - \frac{\$100}{0.02(1.02)^{30}} \approx \$5,000 - \$2760.35 \approx \$2,239.65$
- when $t=60$, $PV_{bond} = \frac{\$100}{0.02} - \frac{\$100}{0.02(1.02)^{60}} \approx \$5,000 - \$1,523.91 \approx \$3,476.09$

For $i = 5\%$, $PV_{consol} = \frac{\$100}{0.05} = \$2,000$.

- when $t=10$, $PV_{bond} = \frac{\$100}{0.05} - \frac{\$100}{0.05(1.05)^{10}} \approx \$2,000 - \$1,227.83 \approx \772.17
- when $t=20$, $PV_{bond} = \frac{\$100}{0.05} - \frac{\$100}{0.05(1.05)^{20}} \approx \$2,000 - \$753.78 \approx \$1,246.22$
- when $t=30$, $PV_{bond} = \frac{\$100}{0.05} - \frac{\$100}{0.05(1.05)^{30}} \approx \$2,000 - \$462.75 \approx \$1,537.25$
- when $t=60$, $PV_{bond} = \frac{\$100}{0.05} - \frac{\$100}{0.05(1.05)^{60}} \approx \$2,000 - \$107.07 \approx \$1,892.93$

Problem 2 Monetary Policy and the Stock Market. Assume the short term real policy rate, current and expected, had been 2% until now. Suppose the Fed decides to tighten monetary policy and increase the short-term policy rate (r_{1t}) from 2% to 3%.

- 1) What happens to stock prices if the change in r_{1t} is expected to be temporary, that is, last for only one period? Assume that expected real dividends do not change. Use equation (14.17).

$$Q_t = \frac{D_{t+1}^e}{(1 + r_{1t} + x)} + \frac{D_{t+2}^e}{(1 + r_{1t} + x)(1 + r_{1t+1}^e + x)} + \dots$$

Answer: If the change in r_{1t} is expected to be temporary, only r_{1t} changes to 3% and all expected future short term policy does not change at 2%, so $r_{1t+1}^e = r_{1t+2}^e = \dots = 2\%$. The current stock price will decrease slightly because of increase in r_{1t} .

- 2) What happens to stock prices if the change in r_{1t} is expected to be permanent, that is, is expected to persist? Assume that expected real dividends do not change. Use equation (14.17).

Answer: If the change in r_{1t} is expected to be permanent, $r_{1t} = r_{1t+1}^e = r_{1t+2}^e = \dots = 3\%$. The current stock price will decrease very much.

- 3) What happens to stock prices today if the increase in the real interest rate, current and expected, reflects an increase in expected future output and expected future dividends?

Answer: The effects on stock prices are unclear. The current stock prices may go up or down, depending on how much the increases in the expected future dividends will be.

EXPECTATIONS, CONSUMPTION, AND INVESTMENT

Problem 1 Individual Saving and Aggregate Capital Accumulation. Suppose that every consumer is born with zero financial wealth and lives for three periods: youth, middle age, and old age. Consumers work in the first two periods and retire in the last one. Their income is \$5 in the first period, \$25 in the second, and \$0 in the last one. Inflation and expected inflation are equal to zero, and so is the real interest rate.

- 1) What is the present discounted value of labor income at the beginning of the first period of life? What is the highest sustainable level of consumption such that consumption is equal in all three periods?

Answer: Since the real interest rate is zero, discount factor is zero. Therefore, the present discounted value of labor income $PV_{income} = \$5 + \$25 = \$30$. If the consumer maintains a constant level of consumption, the highest sustainable level is $\frac{\$30}{3} = \10 for each period.

- 2) For each age group, what is the amount of saving that allows consumers to maintain the constant level consumption you found in 1)? (*Hint: Saving can be a negative number if the consumer needs to borrow to maintain a certain level of consumption.*)

Answer: Note that saving is equal to income minus consumption. Therefore, for each age group,

- youth group saves $\$5 - \$10 = -\$5$.
- middle-age group saves $\$25 - \$10 = \$15$.
- old-age group saves $\$0 - \$10 = -\$10$.

- 3) Suppose there are n people born each period. What is total saving in the economy? (*Hint: Add up the saving of each age group. Remember that some age groups may have negative saving.*) Explain.

Answer: Total saving in this economy is $n(-5 + 15 - 10) = 0$.

- 4) What is total financial wealth in the economy? (*Hint: Compute the financial wealth of people at the beginning of the first period of life, the beginning of the second life, and the beginning of the third period. Add the three numbers. Remember that people can be in debt, so financial wealth can be negative.*)

Answer: Recall that every consumer is born with zero financial wealth. Thus, at the beginning of the first period of life, the financial wealth is 0. At the beginning of the second period of life, people have 5 amounts of debt (-5 amount of saving), leading to $-5n$ of financial wealth in the economy. At the beginning of the third period of period, people have 15 amounts of saving. Combining with carry-on debt from the second period, $-5n + 15n = 10n$ of financial wealth at the beginning of the third period. In total, $0 - 5n + 10n = 5n$.

Problem 2 Borrowing Constraints and Aggregate Capital Accumulation. Continue with the setup from Problem 1, but suppose now that borrowing constraint do not allow young consumers to borrow. If we call the sum of income and total financial wealth "*cash on hand*", then the borrowing restriction means that consumers cannot consume more than their cash on hand. In each age group, consumers compute their total wealth and then determine their desired level of consumption as the highest level that allows their consumption to be equal in all three periods. However, if at any time, desired

consumption exceeds cash on hand, then consumers are constrained to consume exactly their cash on hand.

- 1) Calculate consumption in each period of life. Compare this answer to 1) of Problem 1 and explain any differences.

Answer: During youth, the desired consumption is $\frac{\$30}{3} = \10 , but cash on hand is only \$5. Thus, consumers consume \$5 during youth. During middle age, the desired consumption of $\frac{\$25}{2} = \12.5 , and people can afford to consume the desired amount of consumption. Consumers save \$12.5 for old age, and they consume \$12.5 during old age. The differences between the answer to 1) stem from the borrowing constraints during youth.

- 2) Calculate total saving for the economy. Compare this answer to your answer to 3) of Problem 1 and explain any differences.

Answer: For each group,

- youth group saves $\$5 - \$5 = \$0$.
- middle-age group saves $\$25 - \$12.5 = \$12.5$.
- old-age group saves $\$0 - \$12.5 = -\$12.5$.

Total saving in this economy is $n(0+12.5-12.5)=0$.

- 3) Derive total financial wealth for the economy. Compare this answer to your answer to 4) of Problem 1 and explain any differences.

Answer: Since every consumer is born with zero financial wealth, the financial wealth at the beginning of the first period of life is still zero. At the beginning of the second period of life, neither debt nor saving. So, financial wealth is also zero at the beginning of the second period of life. At the beginning of the third period of life, there exist carry-on saving from the second period of life by 12.5. So, in total, $0 + 0 + 12.5n = 12.5n$ of financial wealth in this economy.

- 4) Consider the following statement: "Financial liberalization may be good for individual consumers, but it is bad for overall capital accumulation." Discuss.

Answer: As we have seen in Problem 1 and Problem 2, financial liberalization can increase individual consumption by allowing people to borrow to consume during the early stage of life. However, it may lead to less financial wealth and, consequently, less capital accumulation.

EXPECTATIONS, OUTPUT, AND POLICY

Problem 1 The New *IS-LM*. For each of the changes in expectations in 1) through 4), determine whether there is a shift in the *IS* curve, the *LM* curve, both curves, or neither. In each case assume that no other exogenous variable is changing.

- 1) a decrease in the expected future real interest rate

Answer: A decrease in the expected future real interest rate leads to an increase in current investment, shifting the *IS* curve rightward.

- 2) an increase in the current real policy interest rate

Answer: The *LM* curve shifts upward.

- 3) an increase in expected future taxes

Answer: The effects of an increase in expected future taxes come into threefold.

- First, an increase in expected future taxes decreases private spending, shifting the IS curve leftward.
- Second, an increase in expected future taxes increases total saving, thus a decrease in expected future real interest rate. Consequently, it shifts the IS curve rightward.
- Lastly, an increase in investment due to a decrease in real interest rate in the medium run leads to an increase in output in the long run. This shifts the IS curve rightward.

The net effect on the IS curve is unclear.

- 4) a decrease in expected future income

Answer: The IS curve shifts leftward.

Problem 2 Effects of Tax Cuts. A new president, who promised during the campaign that she would cut taxes, has just been elected. People trust that she will keep her promise, but expect that the tax cuts will be implemented only in the future. Determine the impact of the election victory on current output, the current interest rate, and current private spending under each of the assumptions in 1) through 3). In each case, indicate what you think will happen to Y^e , r^e , and T^e , and then how these changes in expectations affect output today.

- 1) The Fed will not change its current policy rate, but does not make a commitment as to the direction of rates in the future.

Answer: As we have seen in Problem 1, a change in expected future taxes is not clear. A decrease in T^e increases private spending, leading to the IS curve shift to the right. However, in the medium run, total saving decreases, leading to r^e to increase, shifting the IS curve leftward. In the long run, Y^e decreases due to a decrease in capital accumulation, shifting IS curve leftward.

- 2) The Fed will act to prevent any change in current and future output.

Answer: To maintain current and future output levels, the Fed is highly likely to increase r^e . The effect on output today is still ambiguous, but more likely to be negative than the case in 1).

- 3) The Fed will not change either then current real policy interest rate or the future real policy interest rate.

Answer: If the Fed does not change any real policy, the IS curve definitely shifts to the right and current output increases.

THE OPEN ECONOMY

Problem 1 Balance of Payments. Consider two fictional economies, one called the domestic country and the other the foreign country. Given the transactions listed in 1) through 7), construct the balance of payments for each country. If necessary, include a statistical discrepancy.

- 1) The domestic country purchase \$100 in oil from the foreign country.
- 2) Foreign tourists spent \$25 on domestic ski slopes.
- 3) Foreign investors were paid \$15 in dividends from their holdings of domestic equities.
- 4) Domestic residents gave \$25 to foreign charities.
- 5) Domestic businesses borrowed \$65 from foreign banks.
- 6) Foreign investors purchased \$15 of domestic government bonds.
- 7) Domestic investors sold \$50 of their holdings of foreign government bonds.

Answer:

Domestic Country

Current Account		
Exports	2) 25 (exports of goods)	
Imports	1) 100 (imports of services)	
Trade balance		-75
Income received	0	
Income paid	3) 15 (dividend payment) + 4) 25 (charity payment)	
Net income		-40
Current account balance		-115
Financial Account		
Net capital transfers	0	
Increase in foreign holdings of domestic assets	5) 65 (domestic security holdings by foreign banks) + 6) 15 (domestic government bonds)	
Increase in domestic holdings of foreign assets	7) -50 (foreign government bonds sale)	
Financial account balance		130
Statistical discrepancy		15

Foreign Country

Current Account		
Exports	1) 100 (exports of services)	
Imports	2) 25 (imports of goods)	
Trade balance		75
Income received	3) 15 (dividend payment) + 4) 25 (charity payment)	
Income paid	0	
Net income		40
Current account balance		115
Financial Account		
Net capital transfers	0	
Increase in foreign holdings of domestic assets	7) -50 (foreign government bonds purchase)	
Increase in domestic holdings of foreign assets	5) 65 (domestic security purchase by foreign banks) + 6) 15 (domestic government bonds purchase)	
Financial account balance		-130
Statistical discrepancy		-15

Correction: Capital transfers are transfers in which the ownership of a financial asset changes from one party to another. Thus, \$25 charity transfer should appear in the "Income paid" item in Current Account, not in "Net capital transfers".

Problem 2 Uncovered Interest Parity. Consider two bonds, one issued in euros (€) in Germany, and one issued in dollars (\$) in the United States. Assume that both government securities are one-year bonds—paying the face value of the bond one year from now. The exchange rate, E , stands at 0.75 euros per dollar.

The face values and prices on the two bonds are given by

	Face Value	Price
United State	\$10,000	\$9,615.38
Germany	€10,000	€9,433.96

- 1) Compute the nominal interest rate on each of the bonds.

Answer:

$$\begin{aligned}
 i_{US} &= \frac{10,000 - 9,615.38}{9,615.38} \\
 &= \frac{384.62}{9,615.38} \\
 &\approx 0.04(4\%)
 \end{aligned}$$

$$\begin{aligned}
 i_{GER} &= \frac{10,000 - 9,433.96}{9,433.96} \\
 &= \frac{566.04}{9,433.96} \\
 &\approx 0.06(6\%)
 \end{aligned}$$

- 2) Compute the expected exchange rate next year consistent with uncovered interest parity.

Answer: Using the approximated interest parity condition,

$$\begin{aligned}
 i_{US} &= i_{GER} - \frac{E^e - E}{E} \\
 \Rightarrow 0.04 &= 0.06 - \frac{E^e - 0.75}{0.75} \\
 \Rightarrow E^e &= 0.765
 \end{aligned}$$

- 3) If you expect the dollar to depreciate relative to the euro, which bond should you buy?

Answer: Note that the uncovered interest parity condition implies that

- if $i_{US} = i_{GER} - \frac{E^e - E}{E}$, it is indifferent to hold a US bond or German bond.
- if $i_{US} > i_{GER} - \frac{E^e - E}{E}$, it is better off to hold a US bond.
- if $i_{US} < i_{GER} - \frac{E^e - E}{E}$, it is better off to hold a German bond.

Depreciation of dollar relative to euro means a decrease in E^e , breaking the interest parity status and now it becomes that $i_{US} < i_{GER} - \frac{E^e - E}{E}$. Thus, you should buy a German bond.

- 4) Assume that you are a US investor and you exchange dollars for euros and purchase the German bond today. One year from now, it turns out that the exchange rate, E , is actually 0.72 (0.72 euros buys one dollar). What is your realized rate of return in dollars compared to the realized rate of return you would have made had you held the US bond?

Answer: Realized rate of return in dollars is $\frac{E(1 + i_{GER})}{E^e} = \frac{0.75(1.06)}{0.72} \approx 1.10$, 10% rate of return.
If I have had held the US bond, I would have made 4% rate of return.

- 5) Are the differences in rates of return in 4) consistent with the uncovered interest parity condition? Why or why not?

Answer: The difference in rates of return in 4) is not consistent with the uncovered interest parity condition. The reason why is because the expected exchange rate consistent with uncovered interest parity is 0.765, but the actual realized future exchange rate is 0.72. To put it simply, international investors expected appreciation of the dollar, but it turned out that the value of the dollar decreased comparatively to the euro. Therefore, holding a German bond is much better off than holding a US bond in this case. Note that the uncovered interest parity condition is about equality of expected returns not equality of actual realized returns.

Problem 3 The Goods Market in an Open Economy. Consider an open economy characterized by the following equations:

$$\begin{aligned}C &= c_0 + c_1(Y - T) \\I &= d_0 + d_1Y \\X &= x_1Y^* \\IM &= m_1Y\end{aligned}$$

The parameters x_1 and m_1 are the propensities to export and import, respectively. Assume that the real exchange rate is fixed at a value of 1 and treat foreign income, Y^* , as fixed. Also assume that taxes are fixed and that government purchases are exogenous (i.e., decided by the government). We explore the effectiveness of changes in G under alternative assumptions about the propensity to import.

- 1) Write the equilibrium condition in the market for domestic goods and solve for Y .

Answer: The equilibrium condition for domestic goods in the open market requires

$$\begin{aligned}Y &= Z \\&= C + I + G + X - IM \\&= c_0 + c_1(Y - T) + d_0 + d_1Y + G + x_1Y^* - m_1Y \\&= (c_1 + d_1 - m_1)Y + [c_0 + d_0 + G + x_1Y^* - c_1T]\end{aligned}$$

Solving for Y ,

$$Y = \frac{1}{1 - (c_1 + d_1 - m_1)}[c_0 + d_0 + G + x_1Y^* - c_1T]$$

- 2) Suppose government purchases increase by one unit. What is the effect on output? Assume that $0 < m_1 < c_1 + d_1 < 1$. Explain why.

Answer: If G increases by one unit, equilibrium output Y increases by $\frac{1}{1 - (c_1 + d_1 - m_1)}$. When $0 < m_1 < c_1 + d_1 < 1$, the increase in equilibrium output is larger than the increase in G .

- 3) How do net exports change when government purchases increase by one unit?

Answer: Recall that net exports $NX = X - IM = x_1Y^* - m_1Y$. Thus,

$$G \uparrow \text{ by 1 unit} \rightarrow Y \uparrow \text{ by } \frac{1}{1 - (c_1 + d_1 - m_1)} \text{ unit} \rightarrow NX \downarrow \text{ by } \frac{m_1}{1 - (c_1 + d_1 - m_1)} \text{ unit}.$$

Now consider two economies, one with $m_1 = 0.5$ and the other with $m_1 = 0.1$. Each country is characterized by $(c_1 + d_1) = 0.6$.

- 4) Suppose one of the economies is much larger than the other. Which economy do you expect to have the larger value of m_1 ? Explain.

Answer: Note that m_1 represents the propensity to import. The smaller country is likely to have a

larger m_1 . This is because larger economies tend to produce a wider variety of goods and, therefore spend more additional units of income on domestic goods than smaller economies do.

- 5) Calculate your answers to 2) and 3) for each economy by substituting the appropriate parameter values.

Answer:

For larger economy,

- government spending multiplier $= \frac{1}{1 - (0.6 - 0.1)} = \frac{1}{0.5} = 2$
- import multiplier $= -\frac{0.1}{1 - (0.6 - 0.1)} = -\frac{0.1}{0.5} = -0.2$

For smaller economy,

- government spending multiplier $= \frac{1}{1 - (0.6 - 0.5)} = \frac{1}{0.9} \approx 1.11$
- import multiplier $= -\frac{0.5}{1 - (0.6 - 0.1)} = -\frac{0.5}{0.9} \approx -0.56$

- 6) In which economy will fiscal policy have a larger effect on output? In which economy will fiscal policy have a larger effect on net exports?

Answer: Fiscal policy will have a larger effect on output in a larger economy and a larger effect (in absolute terms) on net exports in a smaller economy.

Problem 4 Exchange Rate Regimes. An overvalued real exchange rate is a rate such that domestic goods are too expensive relative to foreign goods, net exports are too small, and by implication the demand for domestic goods is too low. This leads to difficult policy choices for the government and central bank. The equations (* denotes a value for the foreign country) that describe the economy are:

The IS curve:

$$Y = Y \left(\frac{\bar{E}P}{P^*}, G, T, i^* - \pi^e, Y^* \right)$$

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The Phillips curve for the domestic and the foreign economy:

$$\text{Domestic Phillips curve : } \pi - \bar{\pi} = \frac{\alpha}{L}(Y - Y_n)$$

$$\text{Foreign Phillips curve : } \pi^* - \bar{\pi}^* = \frac{\alpha^*}{L^*}(Y^* - Y_n^*)$$

- 1) We assume that the foreign economy is always in medium-run equilibrium. What are the implications of that assumption for foreign output and foreign inflation?

Answer: The given assumption implies that $\pi^* = \bar{\pi}^*$ and $Y^* = Y_n^*$ always hold.

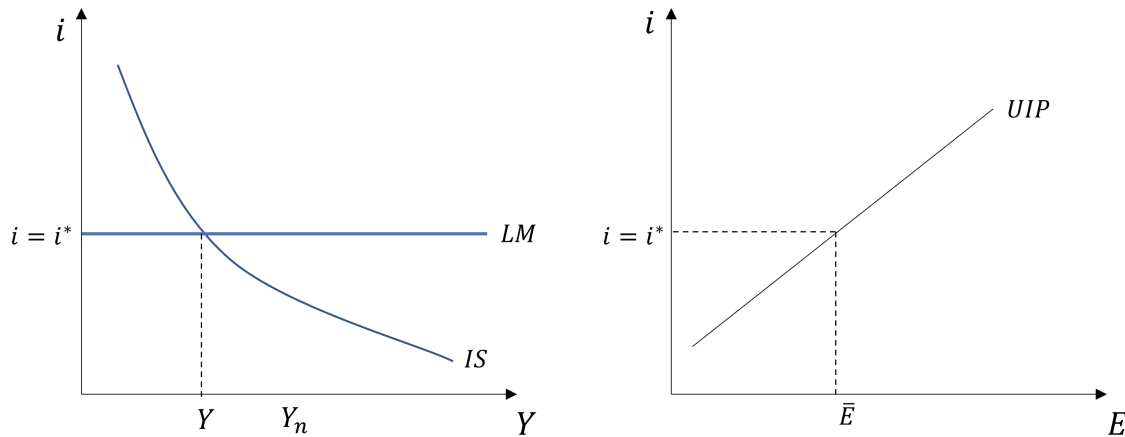
- 2) We assume that the domestic and foreign economies share the same anchored value for the level of expected inflation denoted so that $\bar{\pi} = \bar{\pi}^*$. What is the implication of that assumption once both the domestic and foreign economies are both in medium-run equilibrium?

Answer: When both countries are in medium-run equilibrium, $\pi = \bar{\pi} = \pi^*$, $Y = Y_n$, and $Y^* = Y_n^*$.

- 3) Draw the $IS - LM - UIP$ diagram for the case where the domestic country has an overvalued nominal exchange rate. What is the key feature of that diagram? Under fixed exchange rates

without a devaluation, how does the economy return to its medium-run equilibrium?

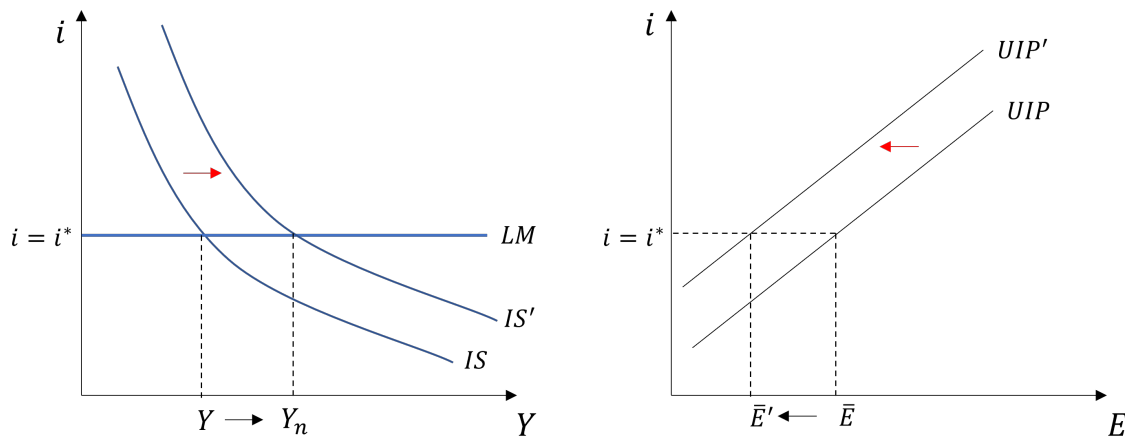
Answer:



With an overvalued nominal exchange rate, $Y < Y_n$ in the short run equilibrium. Note that $\pi < \bar{\pi} = \pi^*$, $P < P^*$ which leads to a depreciation of real exchange rate, even though nominal exchange rate is fixed at $\bar{E}$. Net exports increase and the domestic economy returns to its medium-run equilibrium without a devaluation of nominal fixed exchange rate.

- 4) Draw the $IS - LM - UIP$ diagram for the case where the domestic country has an overvalued nominal exchange rate. Show how the economy can return to its medium-run equilibrium when a devaluation is a policy choice.

Answer:



With a devaluation of fixed nominal exchange rate, the UIP curve shifts to the left given interest rate. Real exchange rate decreases and net exports increase, shifting the IS curve to the right and the domestic economy can return to its medium-run equilibrium.

- 5) Recall that the assumption has been made that interest rate parity holds, so $i = i^*$ at all times. Compare the returns on the domestic bond and the returns on the foreign bond in the period of the devaluation. Will bond holders continue to believe there is a completely fixed nominal exchange rate? If bond holders believe that another devaluation is possible, what are the consequences for domestic interest rates?

Answer: Note that the uncovered interest parity condition says $i = i^* - \frac{E^e - E}{E}$. In the period of the devaluation, the expected appreciation term decreases, leading to $i < i^* - \frac{E^e - E}{E}$. Bond

holders will not continue to believe there is a completely fixed nominal exchange rate. If bond holders expect another devaluation is possible, domestic interest rates must increase a lot to satisfy the uncovered interest parity. If the domestic economy cannot make it, a large amount of capital outflow will occur, leading to an exchange crisis.

BACK TO POLICY

Problem 1 Deficit Reduction as a Prisoner's Dilemma Game. Suppose there is a budget deficit. It can be reduced by cutting military spending, by cutting welfare programs, or by cutting both. The Democrats have to decide whether to support cuts in welfare programs. The Republicans have to decide whether to support cuts in military spending. The possible outcomes are represented in the following table:

(A game with a payoff structure like the one in this problem, and that produces the outcome, is known as a prisoner's dilemma.)

		Welfare Cuts	
		Yes	No
Defense Cuts	Yes	(R=1,D=-2)	(R=-2,D=3)
	No	(R=3,D=-2)	(R=-1,D=-1)

The table presents payoffs to each party under the various outcomes. Think of a payoff as a measure of happiness for a given party under a given outcome. If Democrats vote for welfare cuts, and Republicans vote against cuts in military spending, the Republicans receive a payoff of 3, and the Democrats receive a payoff of -2 .

- 1) If the Republicans decide to cut military spending, what is the best response of the Democrats? Given this response, what is the payoff for the Republicans?

Answer: If the Republicans decided to cut military spending, the best response of the Democrats is not to cut welfare program. In this case, the payoff for the Republicans is -2 .

- 2) If the Republicans decided not to cut military spending, what is the best response of the Democrats? Given this response, what is the payoff for the Republicans?

Answer: If the Republicans decided not to cut military spending, the best response for the Democrats is not to cut the welfare program. In this case, the payoff for the Republicans is -1 . Note that the Democrats choose "No," no matter what the Republicans choose. Such strategy is called *strictly dominant strategy*, which gives a higher payoff given any strategy of the other player.

- 3) What will the Republicans do? What will the Democrats do? Will the budget deficit be reduced? Why or why not? Is there a way to improve the outcome?

Answer: The payoff of the Republicans when they choose "No" is larger than the one when "Yes". Thus, the Republicans choose "No". Predicting the Republicans' action, the Democrats choose "No" as well. The budget deficit will not be reduced. Definitely, this outcome is not desirable for a whole society.

If we transform the structure of the game into a cooperative game, they might come up with several axioms and split the social surplus, leading to a better outcome.

Also, according to *folk theorem*, if players are patient enough, any feasible and individually rational payoff can be supported as an equilibrium in an infinite repeated game. Note that a binding penalty needs to be introduced when one of the player does not follow cooperation.